

INT332:DEVOPS VIRTUALIZATION AND CONFIGURATION MANAGEMENT

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Explain DevOps infrastructure concepts including containerization, runtimes, process isolation, namespaces, and resource management mechanisms.

CO2 :: Apply Docker concepts to build, manage, and distribute container images using networking, storage, and registries.

CO3 :: Construct microservices applications using Docker Compose with multi-container services and dependency configuration.

CO4 :: Analyze and implement automated builds using Maven through lifecycle management, dependency control, plugins, and Docker integration

CO5 :: Apply Continuous Integration workflows using GitHub Actions for automated builds, testing, image creation, and deployment.

CO6 :: Develop end-to-end CI/CD pipelines using Jenkins integrating source control, build tools, containers, and deployments.

Unit I

Basics of DevOps Infrastructure : Docker daemon, Docker CLI, Docker registry & Hub, Object types: container, image, network, volume, Docker layering & filesystem, Introduction to containers (Origin of containers, Emergence of Modern Containerization and Integration into DevOps), Container runtime, Process isolation & namespaces, Control Groups (cgroups) for resource limits, Container images & layers, Image registries & distribution, Introduction to Docker, Docker Architecture

Unit II

Image Building & Container Management : Docker file Core Concepts: Image layering, Build context & .dockerignore, Dockerfile writing, Basic instructions (FROM, RUN, COPY, ADD, CMD, ENTRYPOINT, WORKDIR, ENV, EXPOSE, VOLUME etc),, Image Creation in Detail: docker build process, Images Tagging /versioning, Inspecting images (history, layers), Docker Networking: Bridge network, Host & overlay networks, DNS inside Docker, Linking containers, Port mapping between host and container, Docker Storage: Volumes vs bind mounts, backing data on host, , Copy-on-write mechanism, Registries: Docker Hub, GitHub Container Registry (GHCR),, Private registries, Authentication & access tokens.

Unit III

Microservices with Docker Compose : Microservices Architecture: Need for microservices, Monolithic vs microservices, Advantages: scalability, isolation, agility, API Gateway., Docker Compose: YAML Structure, Writing docker-compose.yml, version, services, volumes, networks,, Environment variables, Secrets and configs, Build vs image fields in YAML, Service dependency ordering., Use Case Deployments: Multi-container apps (database + backend + frontend), WordPress + MySQL, Node.js + MongoDB, Java Spring Boot + PostgreSQL

Unit IV

Maven Build Automation : Why build tools exist, Problems solved by automated builds, Project Object Model (POM), Directory structure, Build lifecycle phases, validate, compile, test, package, verify, install, deploy, Parent POM, Dependency scope, Transitive dependencies, Version conflicts & resolution, Using dependency Management, Maven Plugins & Execution: Compiler plugin, Surefire plugin (unit testing), Shade plugin (uber jar), Maven wrapper (mvnw), Maven and Docker Integration: dockerfile-maven-plugin, Dockerizing Maven-based applications, Pushing artifacts to registries

Unit V

Continuous Integration (CI) with GitHub Actions : Understanding workflow automation, Events, triggers, workflow directory structure, Key components: workflows, Key components: jobs, steps, actions, runners, Workflow triggers: push, pull request, schedule, , manual workflow, Jobs & matrix strategies, Steps & shell commands, Using marketplace actions, Language-specific actions, Using caching for faster builds, Multi-job workflows, Deploying to servers/cloud using Actions, Runners: GitHub-hosted runners, Self-hosted runners ,Runner security & management, Docker & GitHub Actions: Building Docker images in CI, Pushing to Docker Hub, Pushing to GitHub Container Registry (GHCR), Deployments to servers/clouds

Unit VI

Unit VI

CI/CD with Jenkins : Jenkins Foundations: Jenkins architecture (Master/Agent model), Installation & UI overview, Plugins management, Security, users, roles, Jenkins Pipelines: Freestyle vs Pipeline jobs, Declarative pipeline syntax, Scripted pipeline syntax, Jenkinsfile structure, Parameters,, environment variables, Jenkins multi-branch pipelines, Pipeline Stages: Checkout code from Git, Build, Test, Package, Post actions, Managing artifacts, Docker and Jenkins Integration: Building Docker images using Jenkins, Docker inside Jenkins agents, Using Docker plugins, Publishing images to Docker Hub/GHCR, Jenkins and GitHub integration, Backup & restore, Pipeline best practices., Jenkins and Maven: Maven installation in Jenkins, Global tool configuration, Running Maven builds in pipelines, Code coverage & test reports, Jenkins CI/CD Deployment Flows: Triggering builds (pollSCM, webhook), Pipeline libraries, Jenkins agents (SSH/SFTP/Container-based), Deployments to servers/clouds

List of Practicals / Experiments:

List of practical.

- Installing Docker on Windows/Linux, Concept of Hyper-V, Basic Docker commands.
- Create a Docker-hub account, Pushing the Docker to container-hub, Saving changes to a Docker container.
- Creating a Docker file
- Types of Docker storage, Linking docker container
- Introduction to micro services
- Docker compose installation, Introduction to YAML files, Writing a Docker Compose file, deploying WordPress using docker compose.
- Introduction to container orchestration
- Creating a Docker swarm cluster, Initializing Docker Swarm create a service in a Docker Swarm.
- Introduction to Docker networks, Creating a service.
- Introduction to continuous integration.
- Jenkins Architecture, Installing Jenkins.
- Managing nodes on Jenkins.
- Understanding CI / CD Pipelines, Creating CI / CD Pipelines.

Text Books:

1. DOCKER DEEP DIVE by NIGEL POULTON, SHROFF/O'REILLY

References:

1. JENKINS CONTINUOUS INTEGRATION COOKBOOK by ALAN MARK BERG, O'REILLY
2. DOCKER: UP AND RUNNING by KARL MATTHIAS, SEAN P. KANE, O'REILLY
3. JENKINS 2 UP AND RUNNING by BRENT LASTER, O'REILLY